SolarSizer AI

solarsizer-ai.vercel.app

Team	SolarSizer. Team lead: Mgbokinyi Chidubem Emmanuel, Awka, Nigeria.
Problem brief	Brief 7: Replace diesel for productive use.
Track and setting	Track 2, Digital Solutions & AI. Setting B, non-residential: small shops and workshops. Also works for homes.
Community and user	Owners of small shops in Nigerian cities who run a petrol generator when the grid fails. The user pays nothing. Installers and pay-as-you-go solar companies would pay per qualified sizing request. The shop pays for its own system, in cash or monthly.
The problem	Grid power is unreliable, so shops burn petrol. Solar can cost less over time, but buyers have no neutral, location-based way to check a quote. Systems end up too small or too big, and nobody shows when solar pays back.
Proposed solution	A web app. You type what you power in plain words. AI turns it into an appliance list you can edit. Tested formulas size the inverter, battery and panels using NASA POWER sun hours for your city, then show cost, payback, a monthly option, and the petrol, CO_2 and money saved. Share the result on WhatsApp. Works offline after the first visit.
Proof of Concept	A shop owner gets a sized, priced system and yearly savings from one plain sentence, on the live site. Next proof step: compare results with real installer quotes.
Brief success metric	Modelled example shop (₦25,000 petrol a week, ₦1,000 a litre, Lagos): about 1,300 litres of petrol and 3.0 tonnes of CO_2 avoided a year, ₦1.3M saved a year, about ₦4.07M kept over 5 years after the system. Assumes solar replaces all generator petrol.
Cost and who pays	Example system: about ₦2.43M (1 kVA, 5 kWh, 3 × 550 W), payback about 23 months, using placeholder prices. Monthly option on example terms: ₦243,000 down, then ₦115,629 a month for 24 months, against ₦108,333 of petrol a month today. That is ₦7,296 more until it ends, so affordability for the lowest-income users depends on real provider terms. At pilot scale the app runs on free tiers of hosting, AI and NASA data; paid tiers at scale are not yet costed.
6–18 month path	Months 0–6: sourced prices from Lagos installers, pilot with 50 shops. Months 6–12: one pay-as-you-go partner accepts SolarSizer results and pays per lead. Months 12–18: all Nigerian states, more installers, monthly price updates.
How it scales	Cities and prices are data files. A new country needs a city list, a local price table and a currency; the sizing maths and NASA data work worldwide. Kenya and Ghana are next candidates.
Biggest risk and test	Wrong or outdated prices give misleading costs. Test: compare results with real installer quotes, then refresh prices monthly from partners. Second risk: the AI misreads appliances; every row is editable and the AI never does the maths.
Prior work	The working app existed before Oct 30, 2026 (tag prior-work-2026-10-29, first pushed Oct 31). During the sprint: Next.js security upgrade, public deployment, live-site testing, an AI model fix, and this submission package.

Sources: NASA POWER (sun hours); IPCC 2006 default of about 2.3 kg CO_2 per litre of petrol. Equipment and petrol prices are placeholder planning estimates.